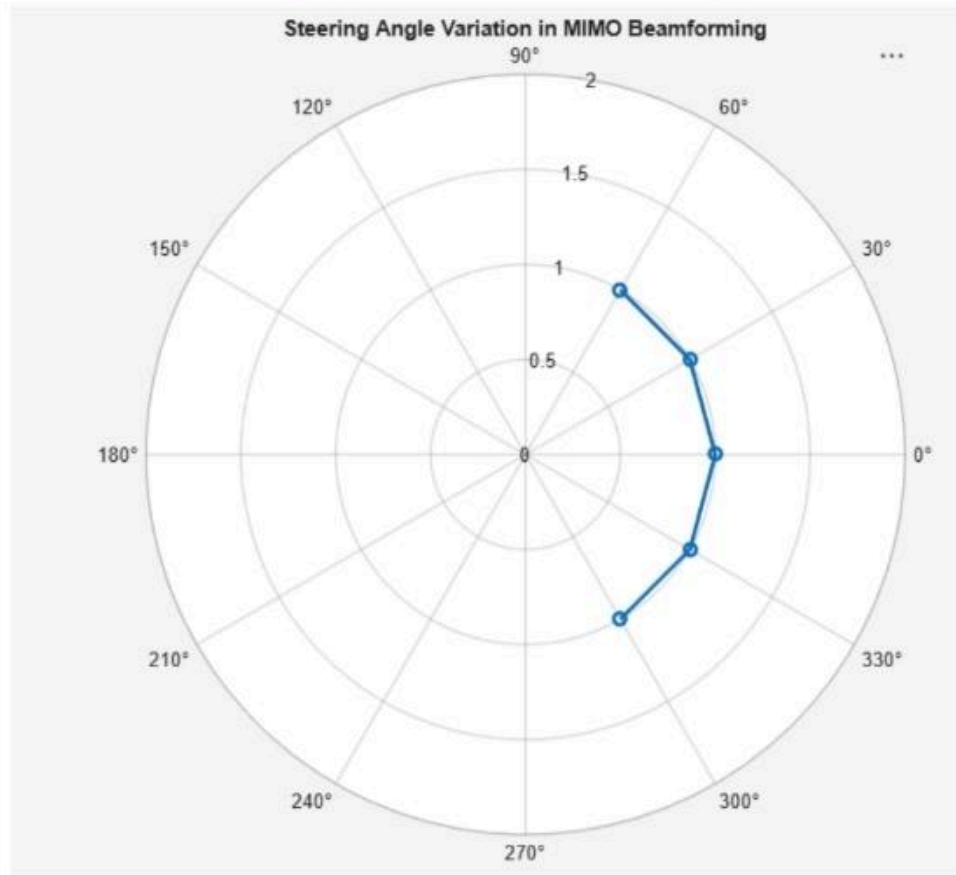


Ex.No.8 Steering Angle Variation in MIMO Beamforming Using MATLAB
Objective 1: Matlab
Code and Output

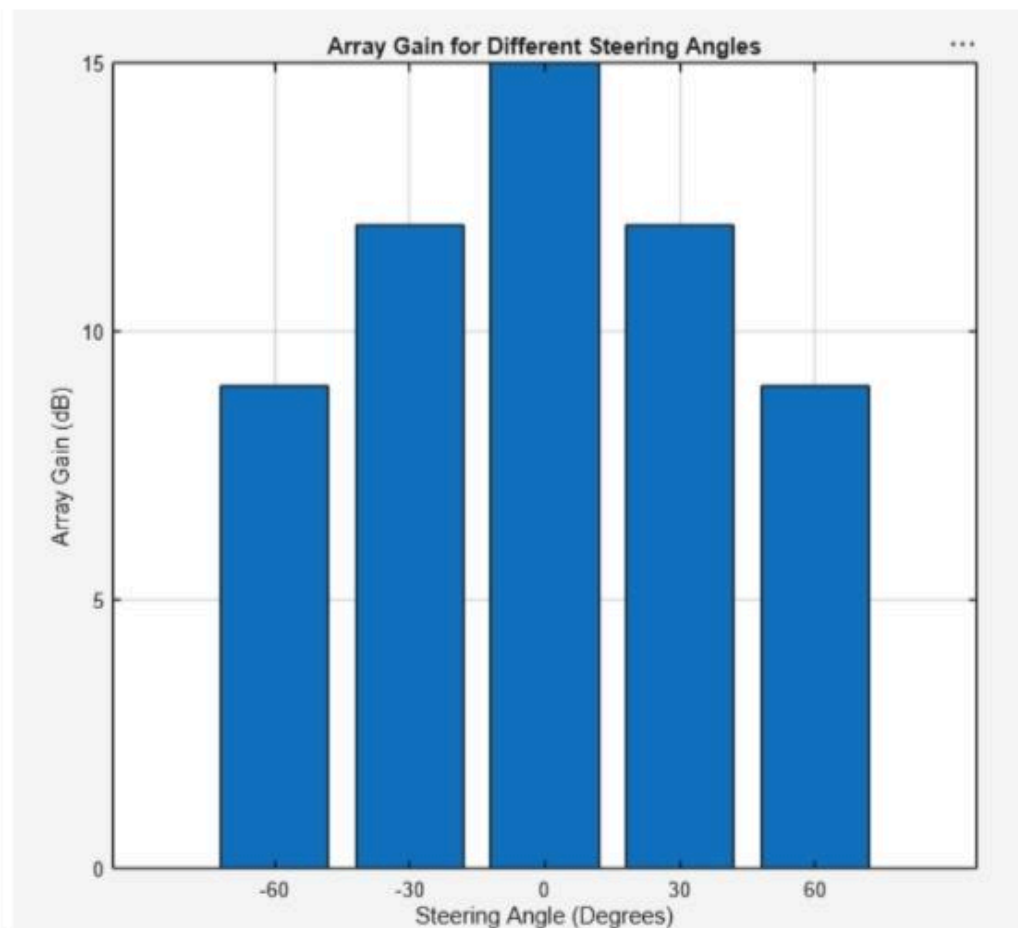
/MATLAB Drive/slotb8.m

```
1  clc;  
2  clear;  
3  close all;  
4  angle = [-60 -30 0 30 60];  
5  gain = ones(size(angle));  
6  figure  
7  % Steering Angles (degrees)  
8  % Normalized Gain  
9  polarplot(deg2rad(angle), gain, 'o-', 'LineWidth', 2)  
10 title('Steering Angle Variation in MIMO Beamforming')
```



Objective 2: Matlab Code and Output

```
12 angle = [-60 -30 0 30 60];  
13 gain = [9 12 15 12 9];  
14 figure  
15 bar(angle, gain)  
16 grid on  
17 % Steering Angles (degrees)  
18 % Array Gain (dB)  
19 xlabel('Steering Angle (Degrees)')  
20 ylabel('Array Gain (dB)')  
21 title('Array Gain for Different Steering Angles')  
22
```



Objective 3: Matlab Code and Output

```
23     desired = [-60 -30 0 30 60];
24     actual = [-58 -32 1 29 62];
25     error = abs(desired - actual);
26     figure
27     % Desired Steering Angle (degrees)
28     % Actual Beam Direction (degrees)
29     % Beam Direction Error
30     plot(desired, error, 'o-', 'LineWidth', 2)
31     grid on
32     xlabel('Desired Steering Angle (Degrees)')
33     ylabel('Beam Direction Error (Degrees)')
34     title('Beam Direction Error for Different Steering Angles')
```

